

Arnav Tiwari

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| [LinkedIn](#)

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| [Portfolio](#)

Professional Summary

Full-Stack Software Engineer with hands-on experience building AI-powered platforms, industrial automation systems, and security dashboards. Delivered measurable impact at Hitachi India (54% reduction in hazardous incidents, 80% faster document workflows) and Payatu Security (50% faster page loads, 40% backend improvement). Specializes in multi-agent LLM orchestration, RESTful API design, and React/Next.js frontends backed by PostgreSQL and MongoDB.

Education

Indian Institute of Information Technology Nagpur (IIIT Nagpur)

November 2022 – July 2026

BTech in Computer Science Engineering | **CGPA: 7.52/10**

Technical Skills

Languages: C, C++, Java, Python, JavaScript, TypeScript

Frontend: React.js, Next.js, Redux, Tailwind CSS

Backend & APIs: Node.js, Express.js, Django, FastAPI, REST, GraphQL, JWT, OAuth 2.0, WebSockets

Testing: Jest, Pytest, Postman

Gen AI: LLM App Dev, RAG, LangChain, LangGraph, Google Gemini API, Prompt Engineering, Multi-agent Orchestration

DevOps & Tooling: Docker, Kubernetes, CI/CD, GitHub Actions, Git, Linux, Bash, Vercel

CS Fundamentals: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

Professional Experience

Software Developer Intern

July 2025 – Present

Hitachi India Pvt Ltd. | Nagpur, Maharashtra

- Engineered SLD visualization modules for steel plant electrical layouts, enabling real-time monitoring of motors and critical equipment — directly contributing to a **33% reduction in motor failures** and **54% reduction in hazardous incidents**.
- Ported Hitachi applications to IntervalZero RTX64 RTOS, eliminating Windows OS dependency and enabling deterministic task scheduling and dynamic memory allocation — improving system efficiency by **43%** and reducing task allocation time by **76%**.
- Built React + Django interfaces for 5 HICADS systems (DPS, RCTMS, PTMS, TTMS, ICQIS), digitalizing manual electrical and steel equipment analysis workflows, reducing operator time consumption by **57%** and improving work efficiency by **43%**.
- Automated CRS document generation pipeline using n8n, reducing turnaround from **2 weeks to 2 days (80% faster)** and enabling delivery of **7–10 project documents weekly** that previously took a month.

Full Stack Developer Intern

May 2025 – July 2025

Payatu Security Consulting Pvt. Ltd. | Pune, Maharashtra

[Certificate](#)

- Built responsive Next.js + SCSS interfaces reducing page load time from **3s to 1.5s (50% faster)**, improving UI maintainability and user experience.
- Developed 15 REST API endpoints and optimized 8 using indexing, query optimization, and caching on a MySQL-backed Node.js backend, improving response time by **40%**.
- Built a real-time logger handling **1,000+ log events/minute** for a 4-role RBAC (Admin, System Engineer, Technician, Employee) WiFi admin dashboard managing **1,000+ users**.

Projects

SkillBridge – AI Adaptive Onboarding Engine

[Link](#) | [GitHub](#)

React, FastAPI, Python, FAISS, spaCy, Sentence-Transformers, NetworkX, Docker, SQLAlchemy

- Built an AI-driven skill gap analyzer processing resumes against job descriptions using a **3-layer matching pipeline** (exact match, skill family, cosine similarity via FAISS) trained on **2,484 resumes** and **2,277 JDs** achieving **skill extraction F1 > 0.82** and **gap detection recall > 85%**.
- Designed an original **Weighted Graph Traversal (WGT) algorithm** with Kahn's topological sort for prerequisite-aware learning pathway generation across a **71-skill taxonomy** and **58-module** curated course catalog.
- Delivered full-stack solution with React DAG visualization (react-flow), real-time 6-stage processing pipeline, LLM reasoning traces, and Docker deployment – generating personalized roadmaps **in under 100 seconds**.

Autopilot – AI Powered Project Planning Platform

[Demo](#) | [GitHub](#)

Next.js, TypeScript, TailwindCSS, Express.js, PostgreSQL, Google Gemini, Groq SDK, OAuth 2.0

- Architected a decoupled, multi-agent AI system with 4 specialized parallel agents (Gemini, Groq SDK) reducing project plan generation from **10–15 minutes to under 3 minutes (83% faster)** vs manual multi-prompt workflows.
- Designed a RESTful API layer with **18 endpoints** orchestrating agent coordination and returning structured execution plans with confidence scores, cost estimations, and task breakdowns.
- Delivered OAuth 2.0 (Google + GitHub) auth, PostgreSQL integration, multi-format export (PDF, DOCX, JSON), i18n, and WCAG-compliant dark-mode UI.

AI-Powered Content Generator Platform

[Demo](#) | [GitHub](#)

Next.js, Clerk, Drizzle ORM, PostgreSQL, Tailwind CSS, TypeScript, Razorpay, Google Gemini API

- Engineered a scalable LLM-powered SaaS platform with **18 dynamic content templates** spanning blog posts, ad copy, social media, and SEO content, reducing content creation time by **70%** vs manual workflows.
- Built PostgreSQL backend with Drizzle ORM featuring schema-level data integrity, query optimization, and full CRUD support for AI generation history and user data.
- Implemented end-to-end authentication via Clerk with usage-based credit tracking, and Razorpay subscription billing with webhook handling for automated plan management.

Achievements

- Rated **5★ (rating 2029)** on CodeChef, ranking among top competitive programmers globally.
- 1st Runner Up – ArtPark Codeforge Hackathon** (Unstop) – competed against teams across India in a software development challenge.
- Achieved **97.6 percentile (AIR 22,007)** in JEE Mains 2022 among 1.1M+ candidates.